



FAIRNESS AWARE MICRO-LENDING RECOMMENDATION SYSTEM

Group 13

- » Aditya Jaini
- » Atanu Choudhury
- » Prateek Grover
- » Rajesh Mattaparthi

Problem Definition

- **What Are We Solving?**

Build an automated micro-lending system that matches global lenders to borrowers without discriminating against marginalized groups.

System Architecture (How It Works)

1. The Recommendation Engine (Scoring & Ranking):

- Matches loan details with lender preferences (like target location or business type).
- Displays the highest-matching loans at the top of the lender's feed.
- **Measured By:** NDCG (evaluates if top matches are actually relevant).

2. The Agentic Component (Fairness Gatekeeper):

- Acts as an automated filter reviewing recommendations before finalizing them.
- Enforces fairness rules so gender or region doesn't restrict loan access.
- **Measured By:** Guardrail Adherence Rate (percentage of decisions that strictly follow fairness rules).

Problem Definition(Continued)

Why It Matters

- **Expands Capital Access:** Helps entrepreneurs without traditional credit histories get funding.
- **Prevents AI Bias:** Blocks algorithms from repeating historical biases against female or rural borrowers.
- **Scales Up Lending:** Capable of automatically processing and matching thousands of global loan applications.

Project Scope

Scope & Limitations

- **In Scope:** Anonymous Kiva crowdfunding data combined with UN economic stats.
- **Out of Scope:** Private credit reports or live banking data.

Success Metrics

High Accuracy: Reach an NDCG ≥ 0.8 so top matches are placed correctly.

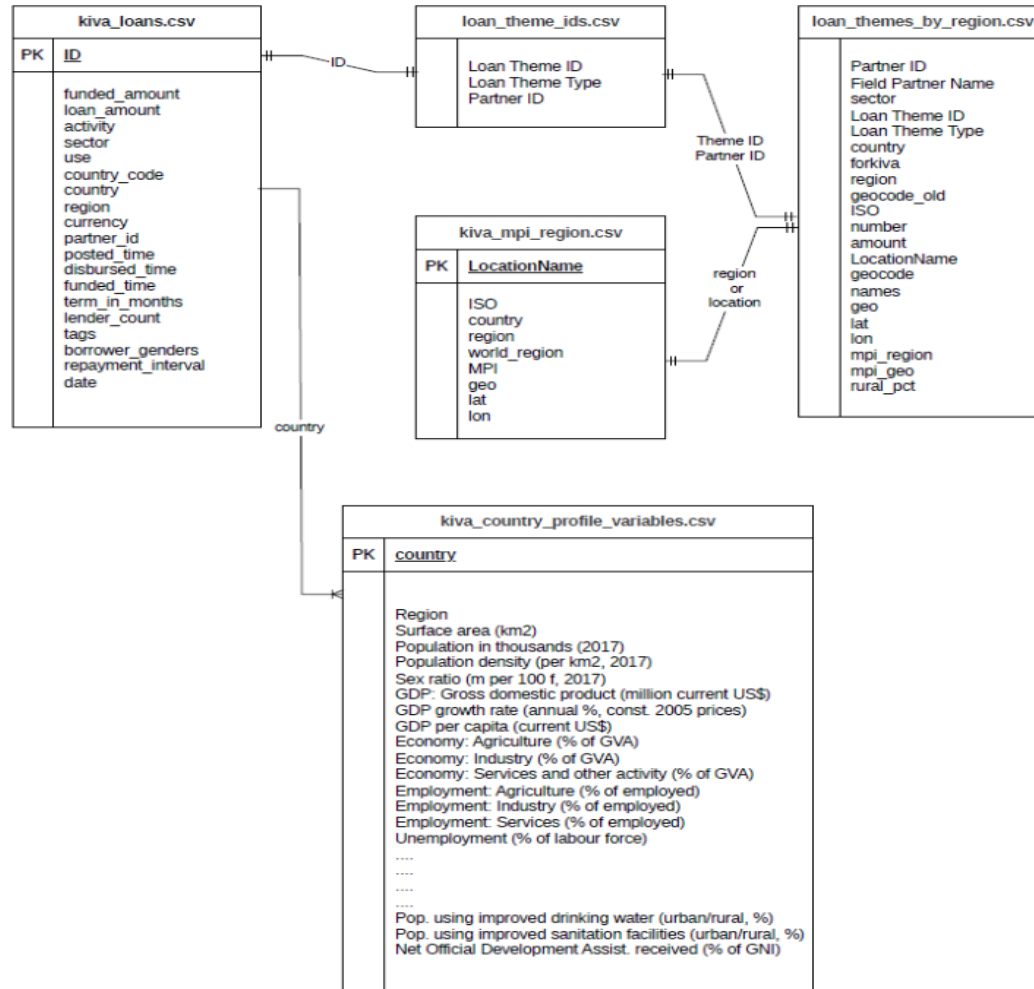
- **Fair Selection:** Keep selection rate differences below 5% across protected groups (gender and region).
- **Zero Violations:** Maintain a 0% Guardrail Violation Rate, meaning no unfair allocations are approved.

Data Sources & Exploration

Dataset	What It Contains	Data Granularity
Loan Transactions	Amounts, business sectors, repayment terms, gender	Individual loan
Regional Data	Partner organizations, locations, rural population %	Local partner region
Poverty Indices	Multidimensional Poverty Index (MPI) scores	State / Province
Partner Mappings	Links loan IDs to specific partners and regions	Unique loan
UN Country Profiles	GDP, population density, national sex ratios	Country level

Data Sources & Exploration(Continued)

Data model - Entity relation



Quick Data Highlights

- **Loan Records:** 671,205 loans analyzed (Agriculture was the largest sector with 180,302 loans).
- **Local Regions:** 15,736 regional operational records.
- **Poverty Scores:** Average poverty (MPI) score of 0.211 across 2,772 regions.
- **Macro Profiles:** 86 country profiles covering 50 economic and social indicators.

Data challenges & recommendations

Challenge	Root Cause	Solution & Mitigation Strategy
Location Inconsistency	Discrepancies in naming across global partners	Standardized all location(country) fields to ISO 3166 alpha-3 codes.
Algorithmic Bias	Standard ranking favors high-volume, low-risk areas	Built active policy guardrails into the AI Agent gatekeeper.
Missing Regional Data	Sparse subnational welfare indices (MPI)	Used proxy imputation based on national economic averages.
Data Pipeline Scale	Heavy compute cost joining multi-layer datasets	Optimized relational joining via primary key mapping.

Model Proposals – Team A

Hybrid GBDT Ranking with Post-Filter Engine

- **Architecture:** Two-layer hybrid framework combining a Learning-to-Rank recommendation model with a deterministic post-filtering policy engine.
- **Layer 1 (Recommendation Engine):** Implements LightGBM (*LGBMRanker* with *lamdarank objective*) on lender-loan cross-features to generate raw compatibility score $S(u,i)$.
- **Layer 2 (Fairness Policy Filter):** Re-ranks top-K recommendations by swapping in high-scoring loans from underrepresented groups if selection rate variance exceeds the $\leq 5\%$ threshold.
- **Suitability:** Handles complex tabular data efficiently, ensures model decision transparency via feature importance, and enforces strict fairness constraints independently of feature learning.
- **Implementation:** Built using lightgbm, scikit-learn, and pandas; runs efficiently on standard multi-core CPU architecture.

Evaluation Approaches – Team A

Proposal#1 – NDCG

- NDCG measures the quality of the ranked recommendations list by considering the higher importance to relevant items that appear in the top of the list. A high NDCG score means the more relevant recommendations are likely to be placed on the top of the ranking list.
- NDCG is a position aware ranking metric that rewards placing the most relevant loans at the top of the recommendation list. It supports graded relevance, allowing loans with different levels of suitability to be ranked appropriately. Since lenders are most likely to review the first few recommendations, NDCG closely aligns with the objectives of our recommendation system.

Proposal#2 - Selection Rate Difference, Guardrail Adherence Rate

- Selection Rate Difference is to measure percentage of applications from each protected group, whether the different demographic groups show similar recommendations or not.
- Guardrail Adherence Rate is important metric as it evaluates whether the ethical policy engine is following the predefined fairness rules throughout this process or not. It will help ensure compliance with fairness policies and identify unfair recommendations and evaluate the effectiveness of the ethical agent

Target is NDCG@10 of at least 0.80, selection rate variance within 5 percent, and a guardrail violation rate of 0

Model Approaches – Team B

Proposal#1

Graph Neural Network & Agentic Guardrail

- **Core Architecture: Bipartite Graph Neural Network (e.g., GraphSAGE)**
 - **Graph Representation:** Models lenders, loans, sectors, and geographic regions as nodes in a bipartite graph.
 - **Relational Embeddings:** Captures multi-hop structural relationships to generate high-dimensional embeddings for recommendations.
- **Fairness Mechanism: Guardrail**
 - Acts as an automated final check on top matches before showing them to users.
 - Re-rank results if any group (gender or region) falls below fairness limits (variance $\geq 5\%$)
- **Best Suited For:** Mapping complex, interconnected data relationships.
- **Implementation:** Built using PyTorch Geometric (PyG), scikit-learn, and pandas; trains efficiently on a single standard GPU or multi-core CPU.

Proposal#2

Learning-to-Rank & Explainable AI

- **Core Architecture: Gradient-Boosted Learning-to-Rank (LightGBM / LGBMRanker)**
 - Uses boosted decision trees to rank loans based on lender preferences and loan details.
 - Handles mixed data types (loan terms, partner history, economic scores) quickly and easily.
- **Fairness & Interpretability: SHAP & Direct In-Model Debiasing**
 - Uses SHAP to show exactly *why* a loan was recommended.
 - Removes bias directly while building the model so no group is penalized.
 - *How it works:* Adjusts decision tree splits and feature weights during training to prevent sensitive factors (like gender or location) from dominating ranks.
- **Best Suited For:** Tabular data, fast live results, and clear explanations.
- **Implementation:** Built using LightGBM, SHAP, scikit-learn, and pandas; runs efficiently on standard multi-core CPU architecture.

Evaluation Approaches – Team B

Part#1

Recommendation Engine Accuracy

- Evaluate the raw predictive power of the scoring model to accurately match lender preferences with loan features before fairness constraints are applied.
- **Metric:** Normalized Discounted Cumulative Gain (NDCG@5).
- **Success Criteria:** Achieve a ranking optimization NDCG score of ≥ 0.8 .

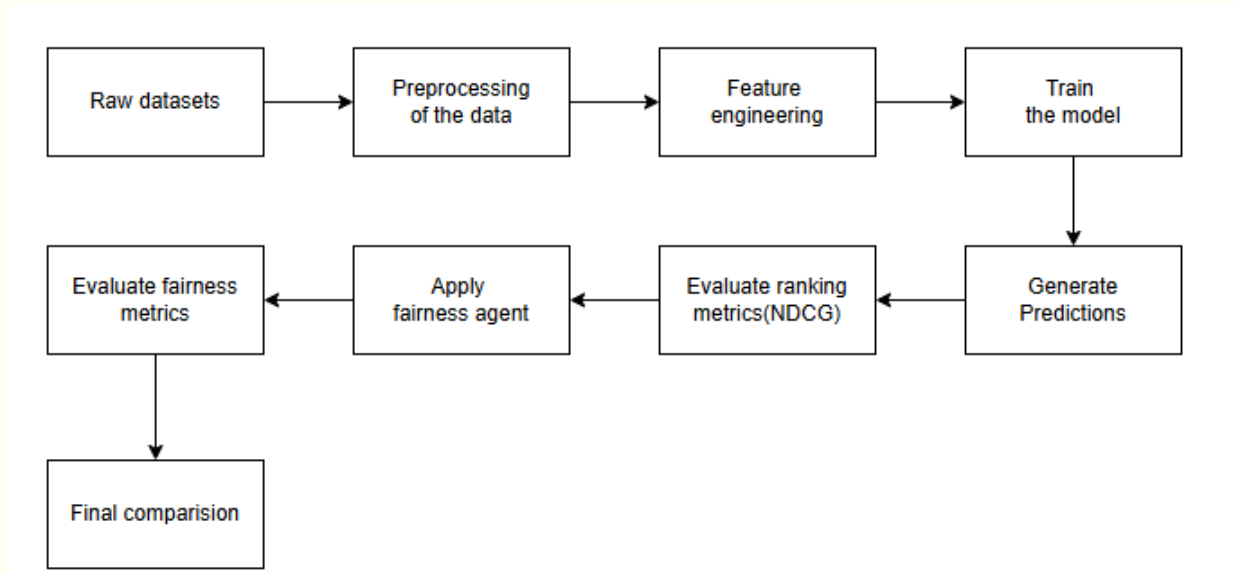
Part#2

System Scalability and Latency

- Evaluate the Agentic Component, which acts as a rules-based gatekeeper to ensure capital allocation does not discriminate against marginalized demographics.
- **Metric** : Selection Rate Variance., Guardrail Adherence Rate / Agentic Guardrail Violation Rate
- **Success Criteria** : selection rate variance of $\leq 5\%$ across protected demographic attributes, specifically gender and region.

Evaluation methodology & scalability

Here is the overview of our evaluation methodology.



- Use evaluation datasets unseen during training or validation to prevent overfitting and performance degradation after deployment.
- Evaluate the Recommendation Engine first, followed by the fairness agent, to improve recommendation quality.
- The final evaluation will assess the entire recommendation pipeline, avoiding misleading conclusions from isolated component testing.
- Since multiple models must be proposed, evaluate them on identical datasets and compare metrics to select the model that best balances quality and fairness.
- Because evaluation requires multiple iterations, the method must be repeatable for consistent results; preprocessing steps and evaluation procedures will be documented.

Evaluation methodology & scalability (Continued)

Scalability and production deployment considerations:

- Since real-world data is much larger, the final production model must handle future dataset growth. To account for this, we selected the Kiva dataset, which contains hundreds of thousands of loans and supplementary data.
- Latency and processing time are key factors for a production-grade solution. We will consider:
 - Model training time for the given dataset
 - How quickly the model generates recommendations
- We aim to minimize additional processing time for fairness evaluation and ethical checks so they do not significantly delay recommendation generation.
- A modular architecture will be used to detect model drift and support targeted retraining without rerunning the entire workflow, saving hours of training time and reducing computational cost. It also supports future enhancements.
- Model performance monitoring will be added to detect drift early and trigger timely retraining.
- CI/CD pipelines will be built using MLOps to reduce production downtime and human errors.

Risks & Mitigation

Category	Risk & Impact	Mitigation Strategy
Data Sparsity & Format Mismatch	Inconsistent country naming across Kiva and UNdata, missing sub-national MPI/spatial metadata	Standardization of all records to ISO 3166 alpha-3 codes
Accuracy and Fairness	Strict post-filtering ($\leq 5\%$ selection rate variance) may reduce Layer 1 LightGBM recommendation scores below target ($NDCG \geq 0.8$)	Finetune Layer 1 <i>lambdarank</i> objectives and establish dynamic score swap-in thresholds in Layer 2 to preserve top-ranked matches
Demographic bias	High-volume and low-risk regions are dominant in the raw data which risks algorithmic bias against female or rural region-based borrowers	Implement deterministic Demographic Parity policy guardrails in Layer 2 to override biased raw scoring before final output
System Scalability	Introduction of possible system processing delay due to the need for re-ranking several thousands of global candidate loans especially during peak system usage time	Usage of table based Gradient Boosted Decision Trees (GBDTs) and rule-based filter consumes less than 10 mins of system time on a standard CPU architecture without the need to consume any additional or significant GPU support

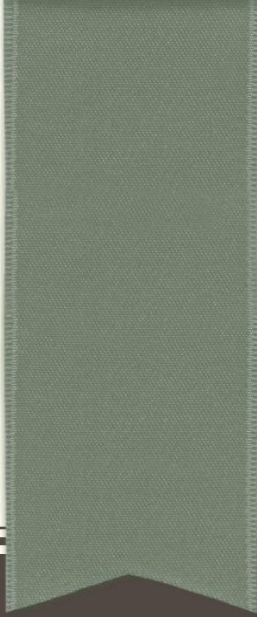
Conclusion

In this project, we aim to tackle an issue with a clear social impact: helping improve financial outcomes for underserved entrepreneurs globally. The challenge is well defined, we know how to measure success, and we have the data needed to build a solid machine learning solution. By merging standard P2P lending recommendations with strict agentic policy constraints, we prove that optimization does not have to come at the cost of social fairness.

From our early look at the data, we can see:

- Everything we need is accessible and ethically collected.
- The data quality is sufficient to build reliable predictions.
- Combining different data sources will work smoothly with proper key standardization.
- The size and scope are appropriate for a graduate-level project.

Overall, this project offers a great balance between technical learning and real-world impact, giving us the chance to strengthen our data science skills while contributing to educational equity, demographic fairness, and student/borrower success.



THANKS FOR WATCHING

Any questions or feedback ? Include in your review comments.
